

Assignment 4 :

Multi-Client Chat Server using TCP

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Objective

Develop a multi-client chat application using TCP sockets where multiple clients communicate through a centralized server. The implementation demonstrates concurrent server design, client management, message broadcasting and performance evaluation.

Network Topology

The experiment was performed using Mininet with one server (h1) and three clients (h2, h3 and h4). Connectivity was verified using nodes, net and pingall.

Software Requirements

Kali Linux, Python 3, Mininet, Wireshark, VirtualBox.

Program Design

The server listens on TCP port 5000, accepts multiple clients using threads, maintains connected users, broadcasts messages and stores logs. The client connects to the server, registers a username and exchanges messages.

Experimental Procedure

1. Create Mininet topology.
2. Verify connectivity.
3. Start server.
4. Connect three clients.
5. Exchange chat messages.

6. Generate chat log.
7. Measure performance.
8. Generate graphs.
9. Capture TCP packets using Wireshark.

Experimental Results

The application successfully supported concurrent communication. Performance results were stored in `performance_results.csv` and graphs were generated for delay and throughput.

Wireshark Verification

Packet captures verified the TCP three-way handshake, chat message transmission, server broadcast packets and TCP connection termination.

Graph Analysis

Average delay increased slightly as the number of clients increased, while throughput remained relatively stable due to efficient multithreading.

Reflection

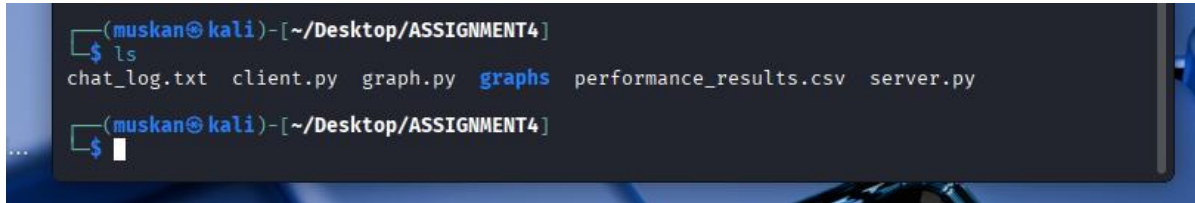
1. Multithreading allows multiple clients to communicate simultaneously.
2. Multiple clients require synchronization and proper resource management.
3. TCP provides reliable delivery using acknowledgements, sequence numbers and retransmissions.
4. Wireshark verified handshake, chat packets, broadcasts and connection termination.

Conclusion

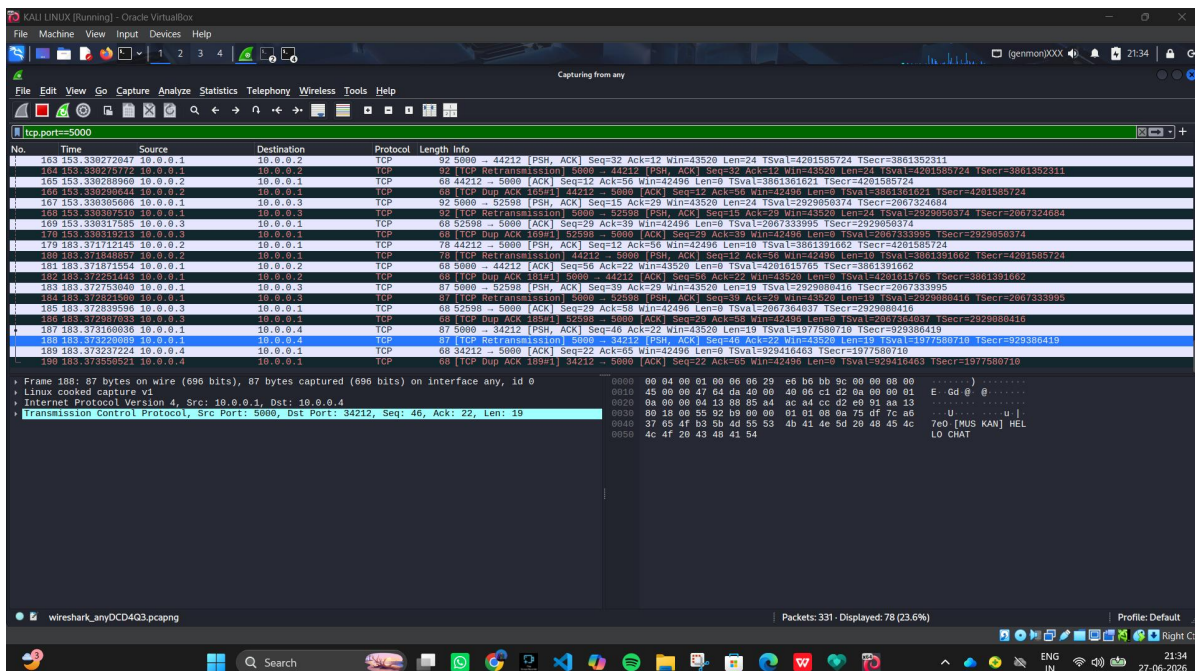
The assignment objectives were successfully achieved. The application demonstrated reliable TCP communication, concurrent client handling and successful packet verification using Wireshark.

Screenshots

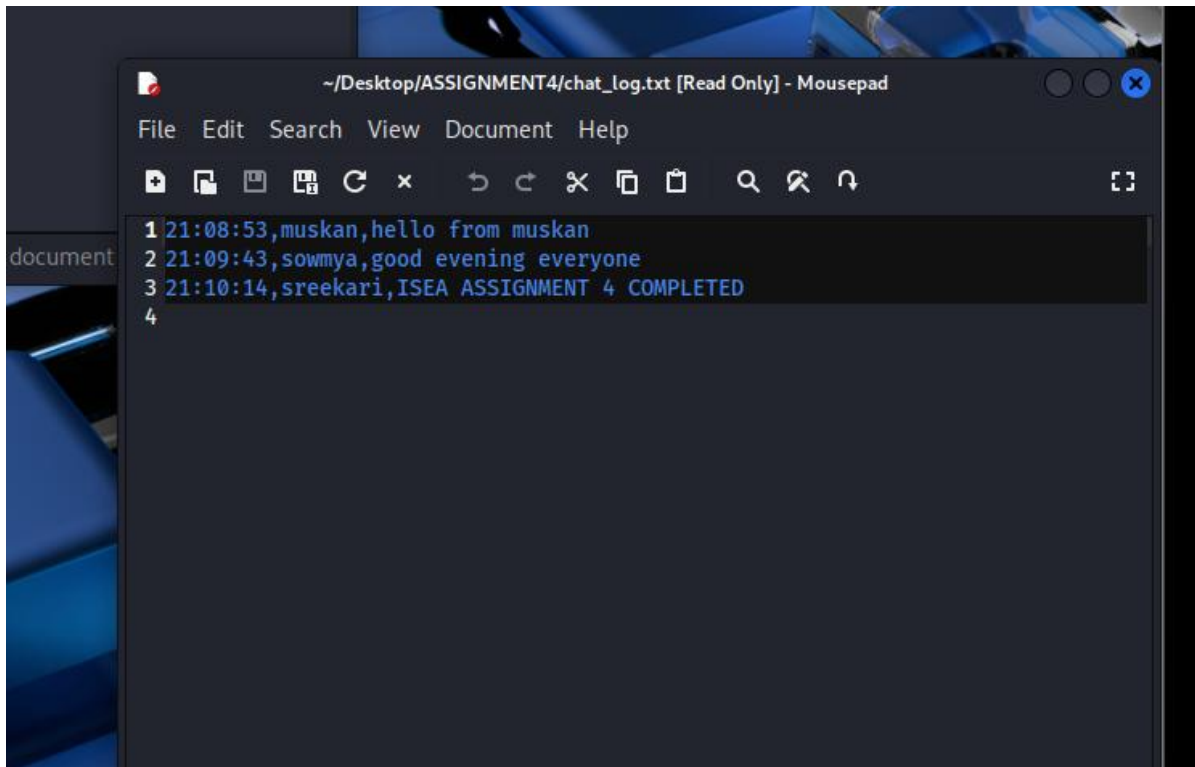
assignment4_files.png



broadcast_message.png



chat_log.png



chat_message.png

KALI LINUX [Running] - Oracle VirtualBox

File Machine View Input Devices Help

muskan@kali: ~/Desktop/ASSIGNMENT4

Session Actions Edit View Help

*any

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 5000

No.	Time	Source	Destination	Protocol	Length	Info
26	71.425561229	10.0.0.2	10.0.0.1	TCP	73	44704 → 5000 [PSH,
27	71.425761099	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission
28	71.425764555	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission
29	71.425766598	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission
30	71.425801012	10.0.0.1	10.0.0.2	TCP	68	5000 → 44704 [ACK]
31	71.426653948	10.0.0.1	10.0.0.2	TCP	68	[TCP Dup ACK 30#1]
32	71.429320525	10.0.0.1	10.0.0.3	TCP	82	5000 → 49706 [PSH,
33	71.429347675	10.0.0.1	10.0.0.4	TCP	82	5000 → 36680 [PSH,
34	71.429464001	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission
35	71.429467410	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission
36	71.429469292	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission
37	71.429494280	10.0.0.3	10.0.0.1	TCP	68	49706 → 5000 [ACK]
38	71.429606081	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission
39	71.429608042	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission
40	71.429608840	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission
41	71.429623523	10.0.0.4	10.0.0.1	TCP	68	36680 → 5000 [ACK]

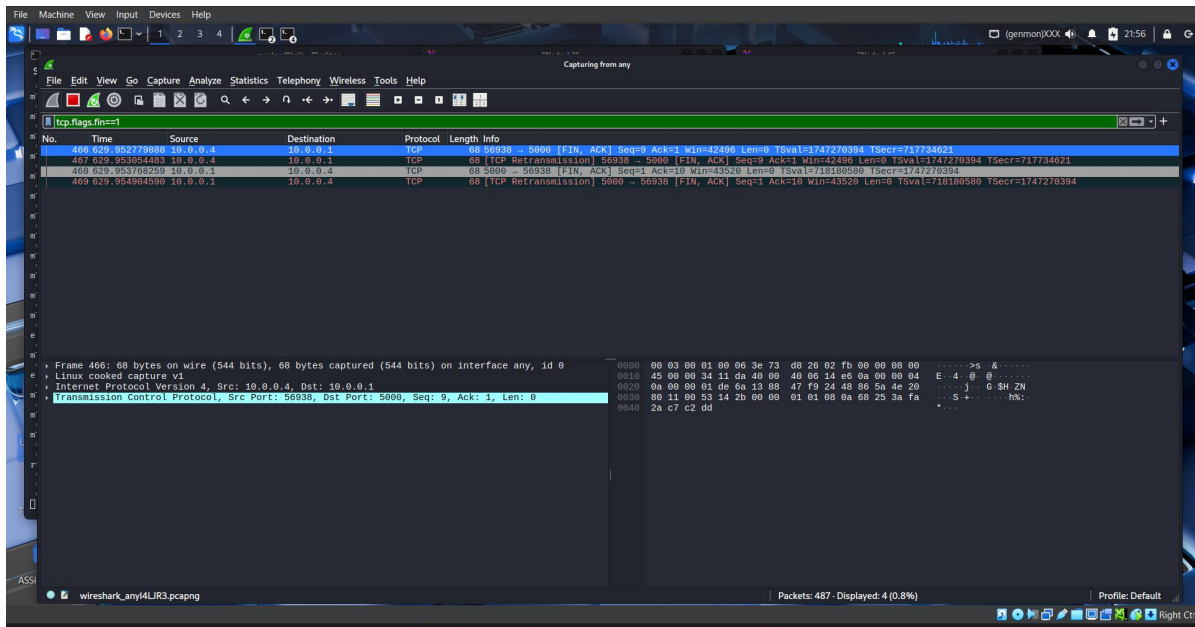
Frame 26: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface eth0
Linux cooked capture v1
Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1
Transmission Control Protocol, Src Port: 44704, Dst Port: 5000
Data (5 bytes)

0000 00 03 00 01 00 06 52 52 ff eb 95 a1 00 00 00 00
0010 45 00 00 39 a2 b4 40 00 40 06 84 08 0a 00 00
0020 0a 00 00 01 ae a0 13 88 f1 48 82 d7 7e 4f bc
0030 80 18 00 53 14 2e 00 00 01 01 08 0a ae 1a 5f
0040 77 a0 84 01 68 65 6c 6c 6f

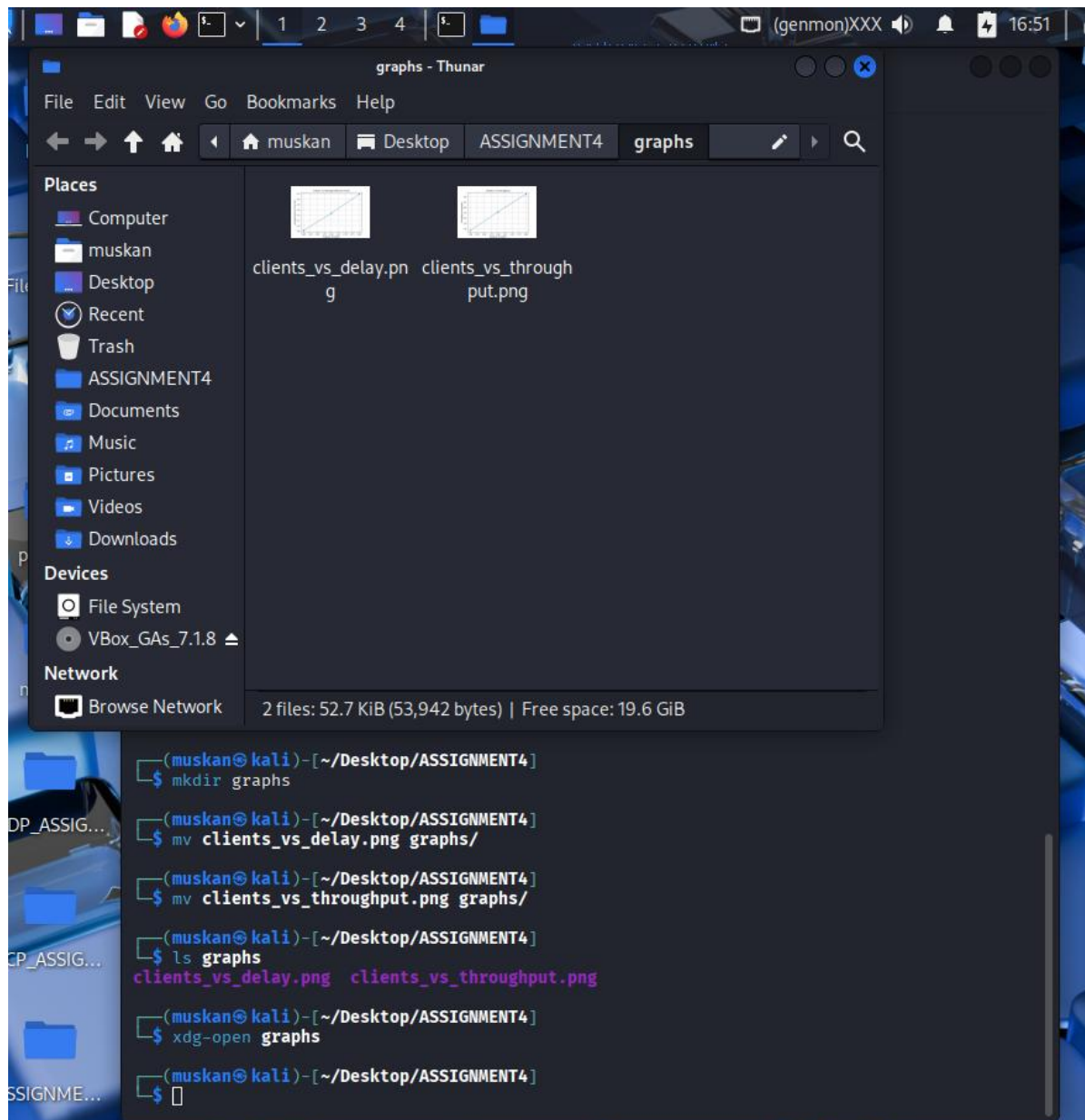
wireshark_anyR2BGR3.pcapng Packets: 461 · Displayed: 270 (58.6%) · Dropped: 0 (0.0%) Profile: Default

Right Ctrl

connection_close.png



graphs_folder.png



h1 ifconfig.png

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet> xterm h1
mininet> xterm h2
mininet> xterm h3
mininet> xterm h4
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
h4 h4-eth0:s1-eth4
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0 s1-eth4:h4-eth0
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::8b3:c8ff:fe86:14f6 prefixlen 64 scopeid 0x20<link>
    ether 0a:b3:c8:86:14:f6 txqueuelen 1000 (Ethernet)
    RX packets 102 bytes 6871 (6.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 57 bytes 3656 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

h2 ifconfig.png

```
mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::cc7b:62ff:fea3:15f6 prefixlen 64 scopeid 0x20<link>
    ether ce:7b:62:a3:15:f6 txqueuelen 1000 (Ethernet)
    RX packets 84 bytes 5950 (5.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 38 bytes 2589 (2.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

h3 ifconfig.png

```
mininet> h3 ifconfig
h3-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.3 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::4035:c4ff:fe3f:b955 prefixlen 64 scopeid 0x20<link>
    ether 42:35:c4:3f:b9:55 txqueuelen 1000 (Ethernet)
    RX packets 82 bytes 5830 (5.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37 bytes 2539 (2.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

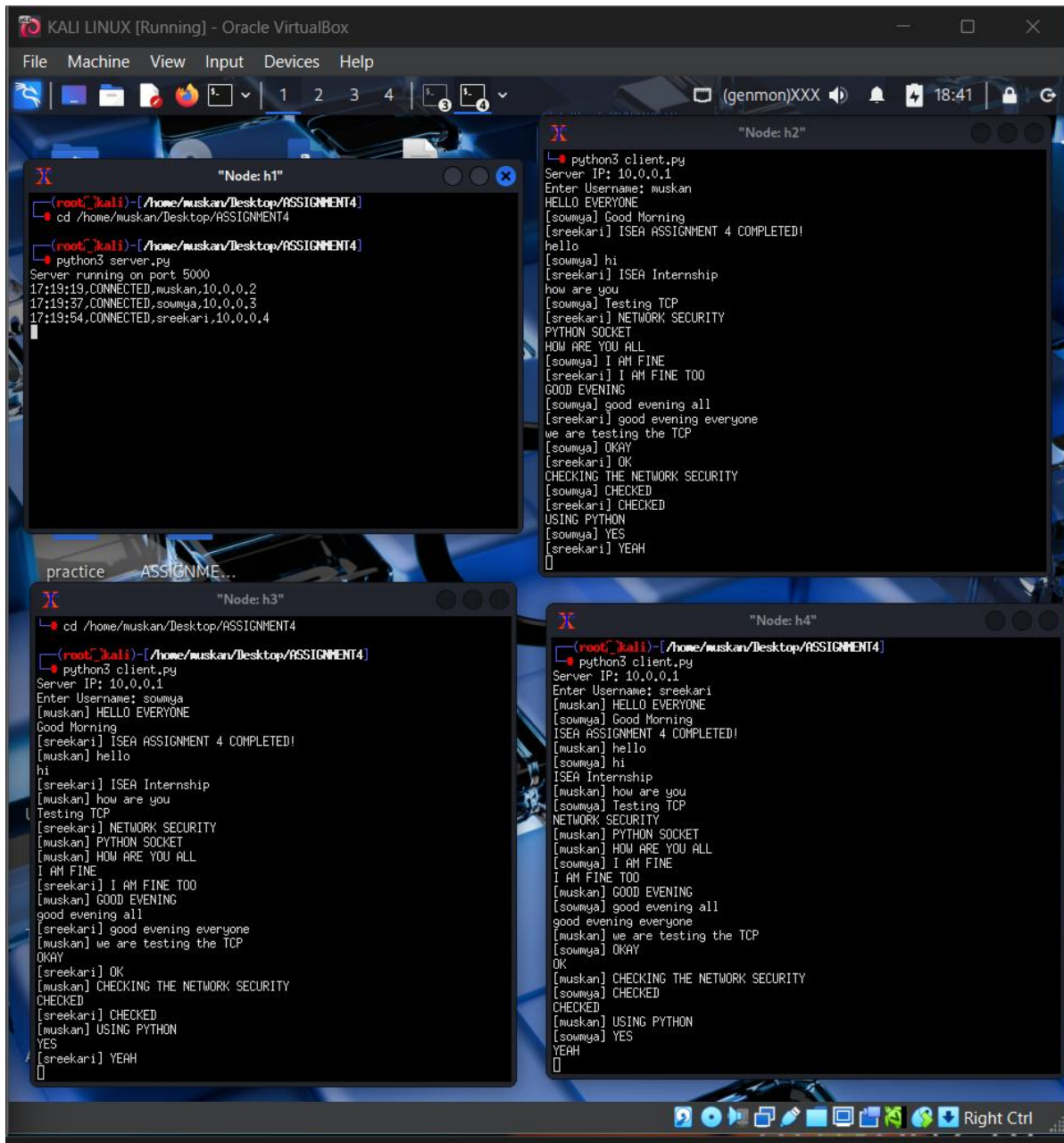
h4 ifconfig.png

```
mininet> h4 ifconfig
h4-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.4 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::c1:c6ff:feb9:7a26 prefixlen 64 scopeid 0x20<link>
    ether 02:c1:c6:b9:7a:26 txqueuelen 1000 (Ethernet)
    RX packets 87 bytes 6156 (6.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 38 bytes 2593 (2.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet> █
```

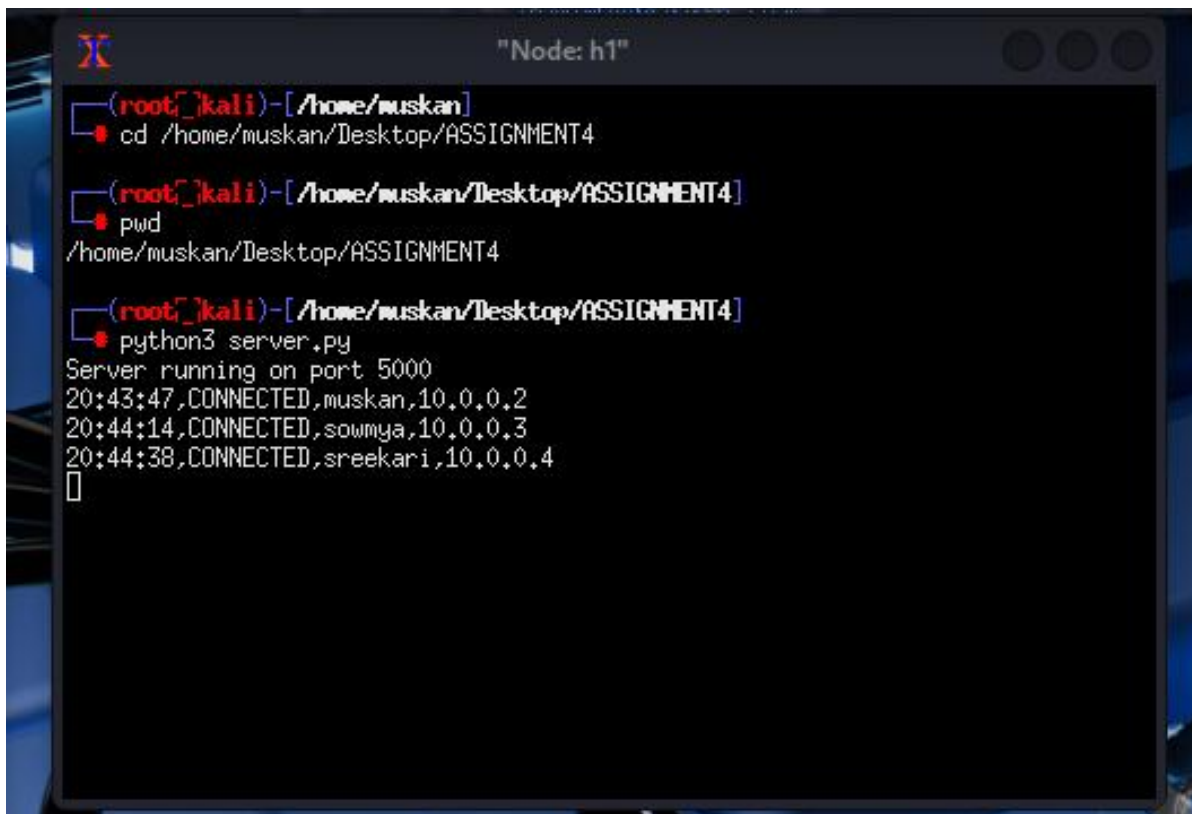
multi_client_chat.png



net.png

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
h4 h4-eth0:s1-eth4
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0 s1-eth4:h4-eth0
mininet> █
```

new h1 .png

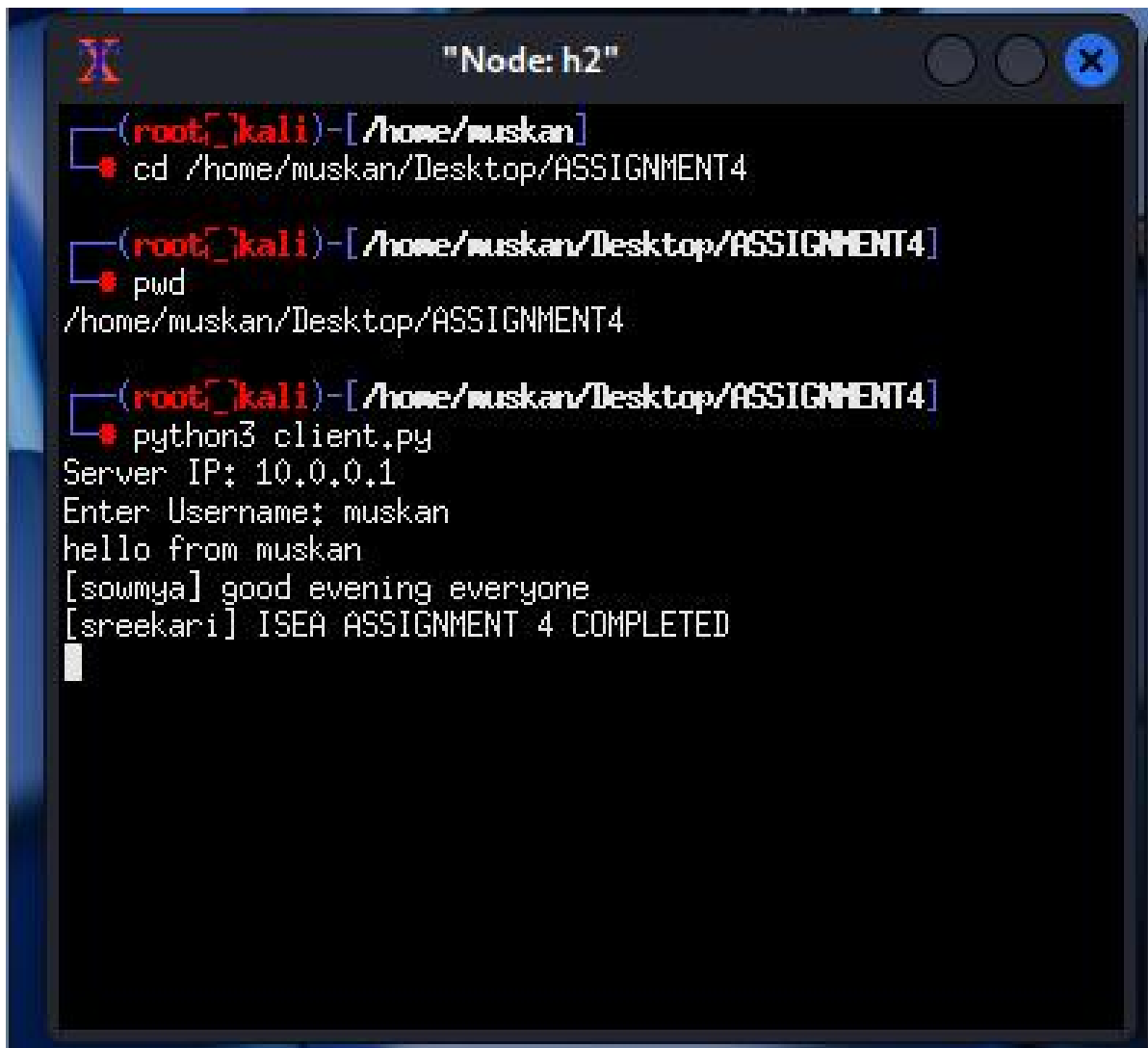


```
(root@kali)~[/home/muskan]
• cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
• pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
• python3 server.py
Server running on port 5000
20:43:47,CONNECTED,muskan,10.0.0.2
20:44:14,CONNECTED,sowmya,10.0.0.3
20:44:38,CONNECTED,sreekari,10.0.0.4
█
```

new h2.png

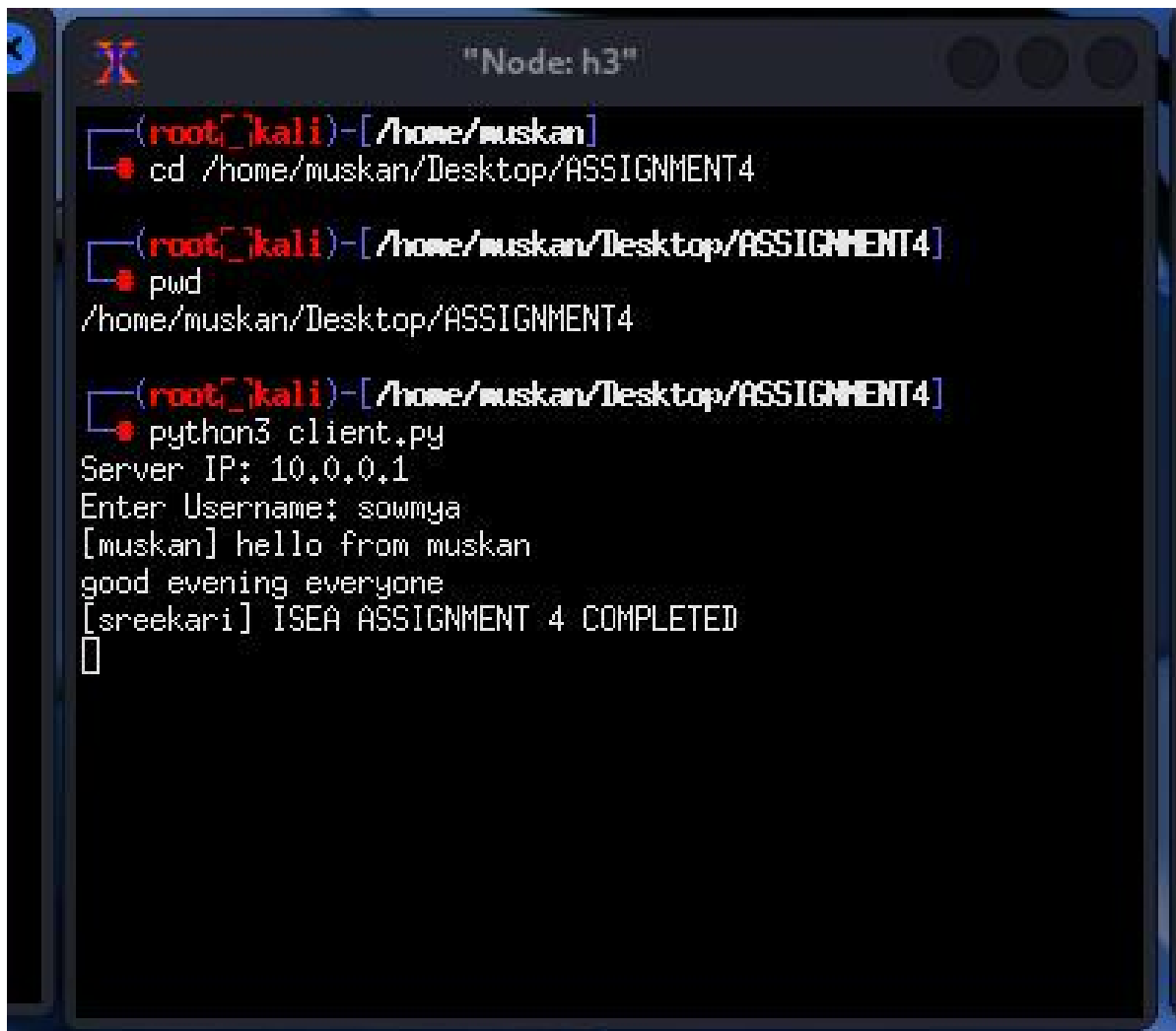


```
(root@kali) - [/home/muskan]
• cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali) - [/home/muskan/Desktop/ASSIGNMENT4]
• pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali) - [/home/muskan/Desktop/ASSIGNMENT4]
• python3 client.py
Server IP: 10.0.0.1
Enter Username: muskan
hello from muskan
[sowmya] good evening everyone
[sreekari] ISEA ASSIGNMENT 4 COMPLETED
```

new h3.png

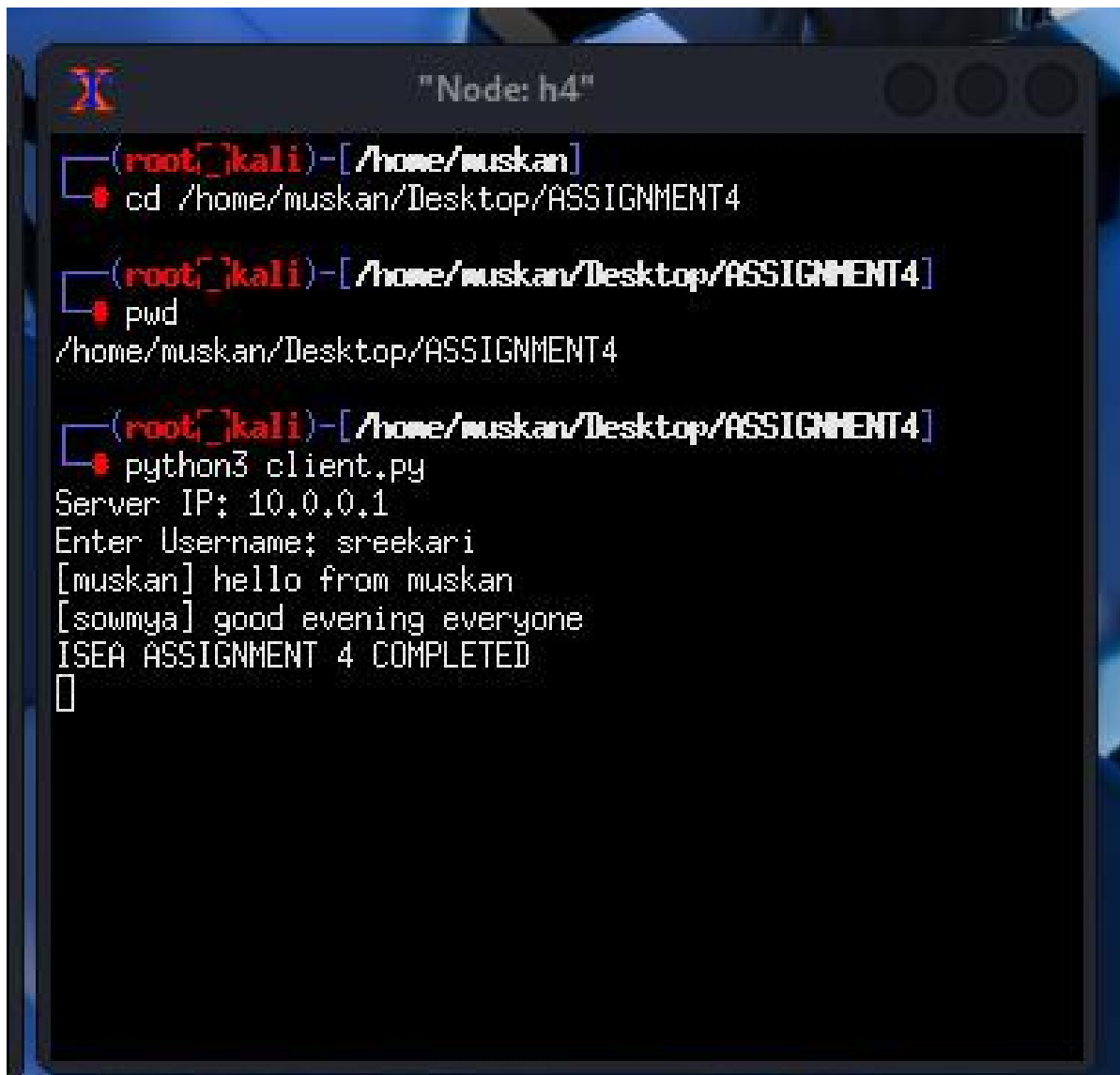


```
(root@kali)-[/home/muskan]
└─$ cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ python3 client.py
Server IP: 10.0.0.1
Enter Username: sowmya
[muskan] hello from muskan
good evening everyone
[sreekari] ISEA ASSIGNMENT 4 COMPLETED
█
```

new h4.png

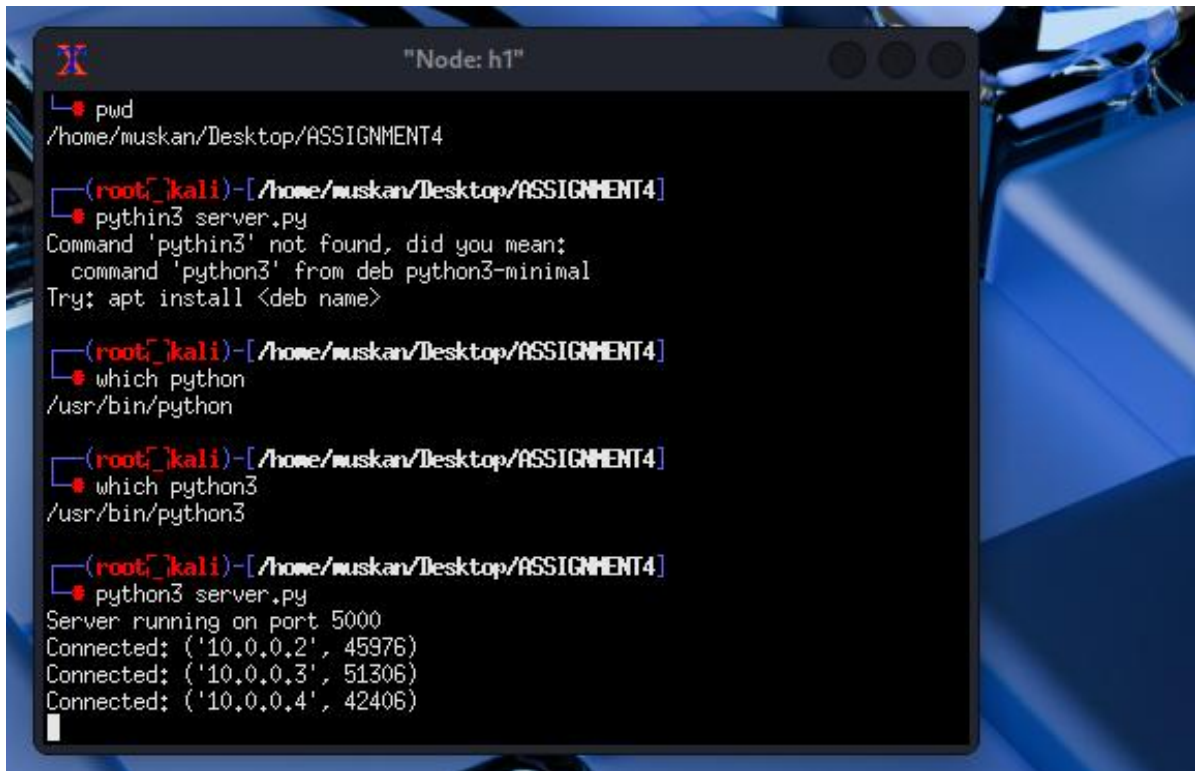


```
(root@kali)-[/home/muskan]
└─$ cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ python3 client.py
Server IP: 10.0.0.1
Enter Username: sreekari
[muskan] hello from muskan
[sowmya] good evening everyone
ISEA ASSIGNMENT 4 COMPLETED
█
```

node h1.png



```
"Node: h1"

• pwd
/home/muskan/Desktop/ASSIGNMENT4

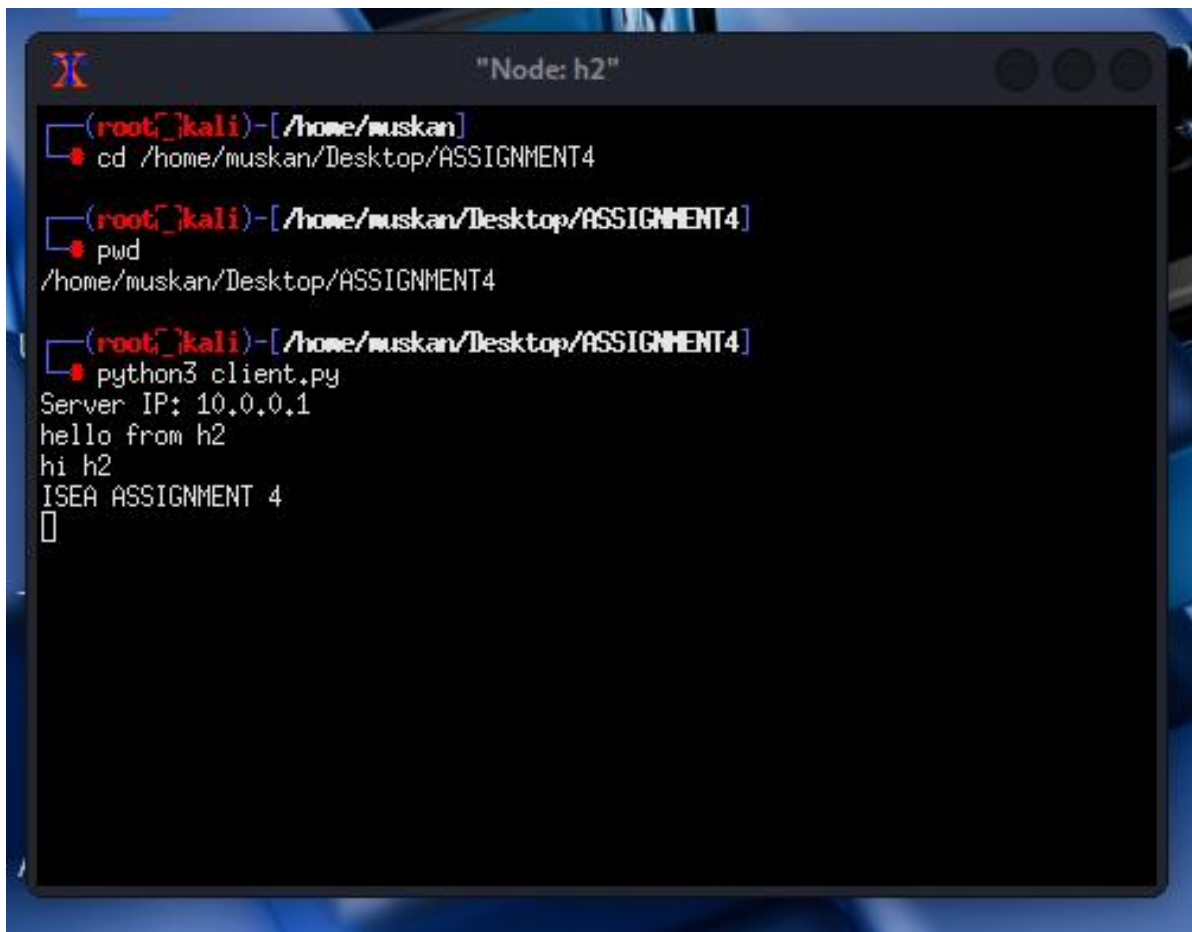
(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
• pythin3 server.py
Command 'pythin3' not found, did you mean:
  command 'python3' from deb python3-minimal
Try: apt install <deb name>

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
• which python
/usr/bin/python

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
• which python3
/usr/bin/python3

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
• python3 server.py
Server running on port 5000
Connected: ('10.0.0.2', 45976)
Connected: ('10.0.0.3', 51306)
Connected: ('10.0.0.4', 42406)
```

node h2.png

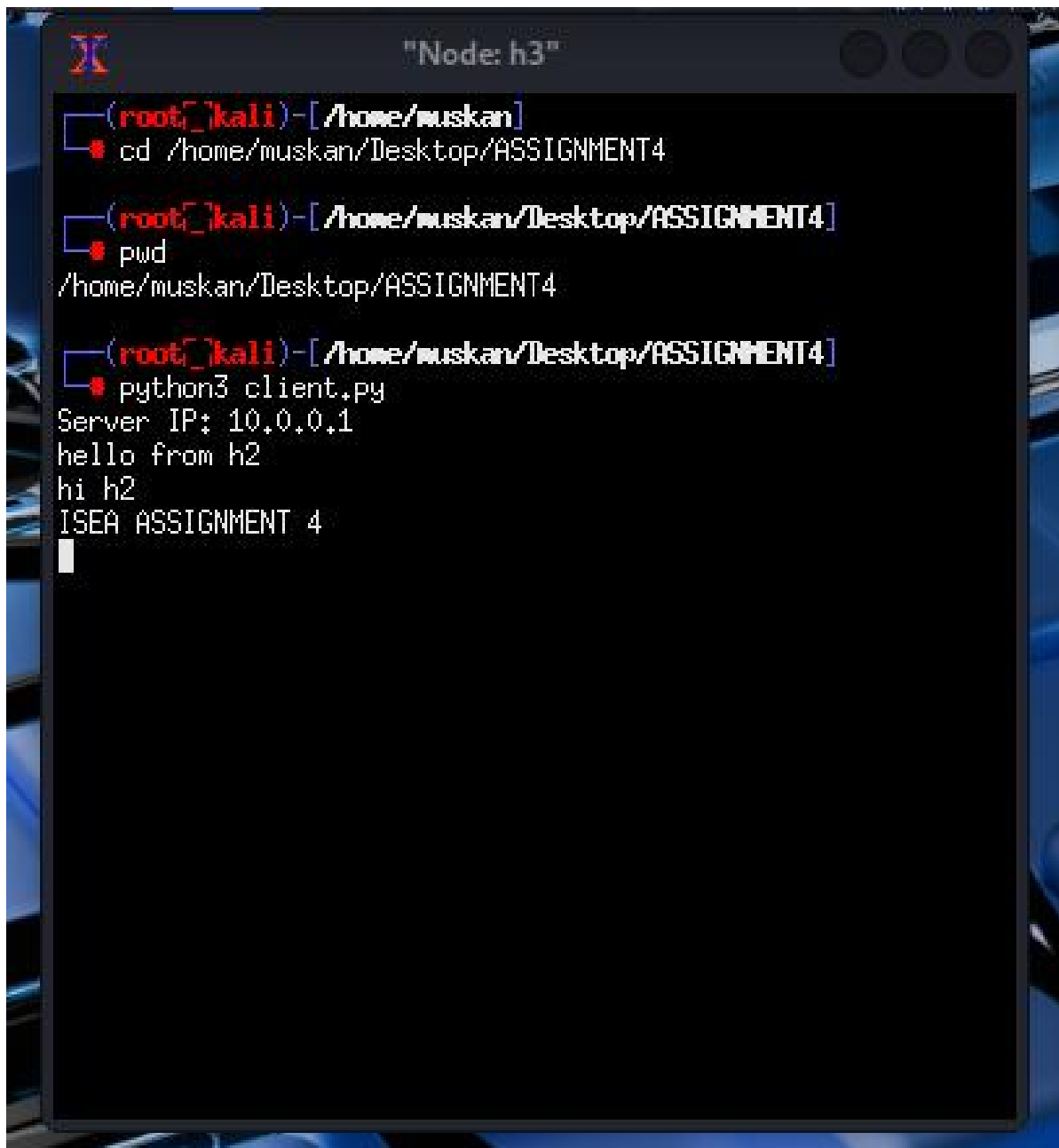


```
(root@kali)~[/home/muskan]
• cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
• pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
• python3 client.py
Server IP: 10.0.0.1
hello from h2
hi h2
ISEA ASSIGNMENT 4
█
```

node h3.png

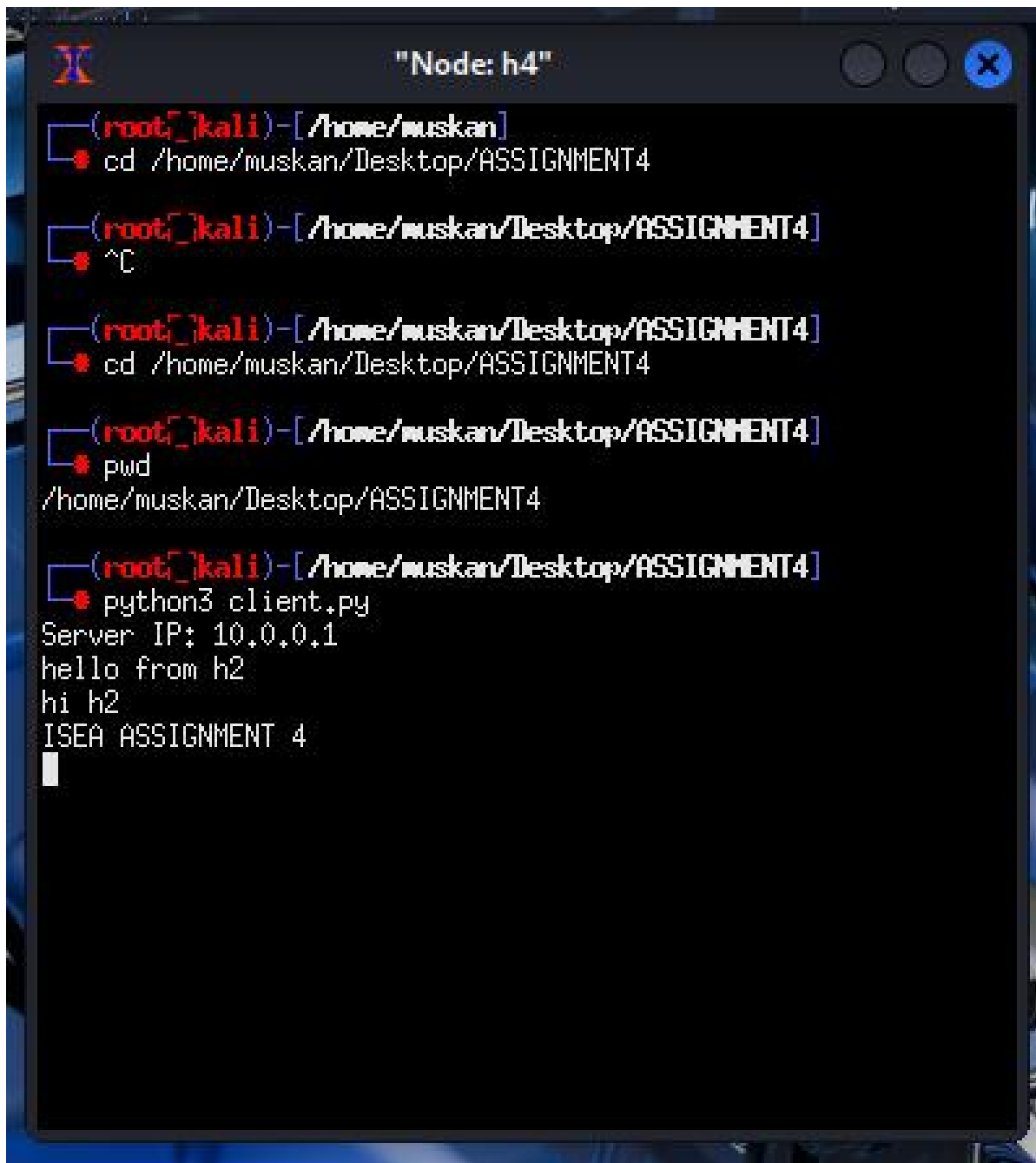


```
(root@kali)-[/home/muskan]
└─$ cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)-[/home/muskan/Desktop/ASSIGNMENT4]
└─$ python3 client.py
Server IP: 10.0.0.1
hello from h2
hi h2
ISEA ASSIGNMENT 4
█
```

node h4.png



```
(root@kali)~[/home/muskan]
* cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
* ^C

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
* cd /home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
* pwd
/home/muskan/Desktop/ASSIGNMENT4

(root@kali)~[/home/muskan/Desktop/ASSIGNMENT4]
* python3 client.py
Server IP: 10.0.0.1
hello from h2
hi h2
ISEA ASSIGNMENT 4
█
```

nodes.png



```
muskan@kali: ~  
Session Actions Edit View Help  
  
(muskan@kali)-[~]  
$ sudo mn --topo single,4  
/usr/local/bin/mn:4: DeprecationWarning: pkg_resources is d  
eprecated as an API. See https://setuptools.pypa.io/en/late  
st/pkg_resources.html  
  import __('pkg_resources').run_script('mininet=2.3.1b4'  
  , 'mn')  
** No default OpenFlow controller found for default switch  
!  
** Falling back to OVS Bridge  
** Creating network  
** Adding controller  
** Adding hosts:  
h1 h2 h3 h4  
** Adding switches:  
s1  
** Adding links:  
(h1, s1) (h2, s1) (h3, s1) (h4, s1)  
** Configuring hosts  
h1 h2 h3 h4  
** Starting controller  
  
** Starting 1 switches  
s1 ...  
** Starting CLI:  
mininet> nodes  
available nodes are:  
h1 h2 h3 h4 s1
```

pingall.png

```
(muskan@kali)-[~]
$ sudo mn --topo single,4
/usr/local/bin/mn:4: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
  __import__('pkg_resources').run_script('mininet=2.3.1b4', 'mn')
*** No default OpenFlow controller found for default switch
!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller

*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 h4 s1
mininet> pingall
*** Ping: testing ping reachability
h1 → h2 h3 h4
h2 → h1 h3 h4
h3 → h1 h2 h4
h4 → h1 h2 h3
*** Results: 0% dropped (12/12 received)
```

tcp_handshake.png

The image shows a Wireshark packet capture of a TCP handshake. The top pane displays a list of packets, with the first three packets (45, 46, and 47) highlighted in blue, indicating they are part of the selected filter `tcp.flags.syn==1`. The middle pane shows the details of the selected packet (No. 46), including the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol layers. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
45	89.13832964	10.0.0.2	10.0.0.1	TCP	76	44212 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=3861297421 TSecr=0 WS=512
46	89.15034692	10.0.0.2	10.0.0.1	TCP	76	(TCP Retransmission) 44212 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=3861297421 TSecr=0 WS=512
47	89.15037410	10.0.0.1	10.0.0.2	TCP	76	5000 → 44212 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=4201521524 TSecr=3861297421 WS=512

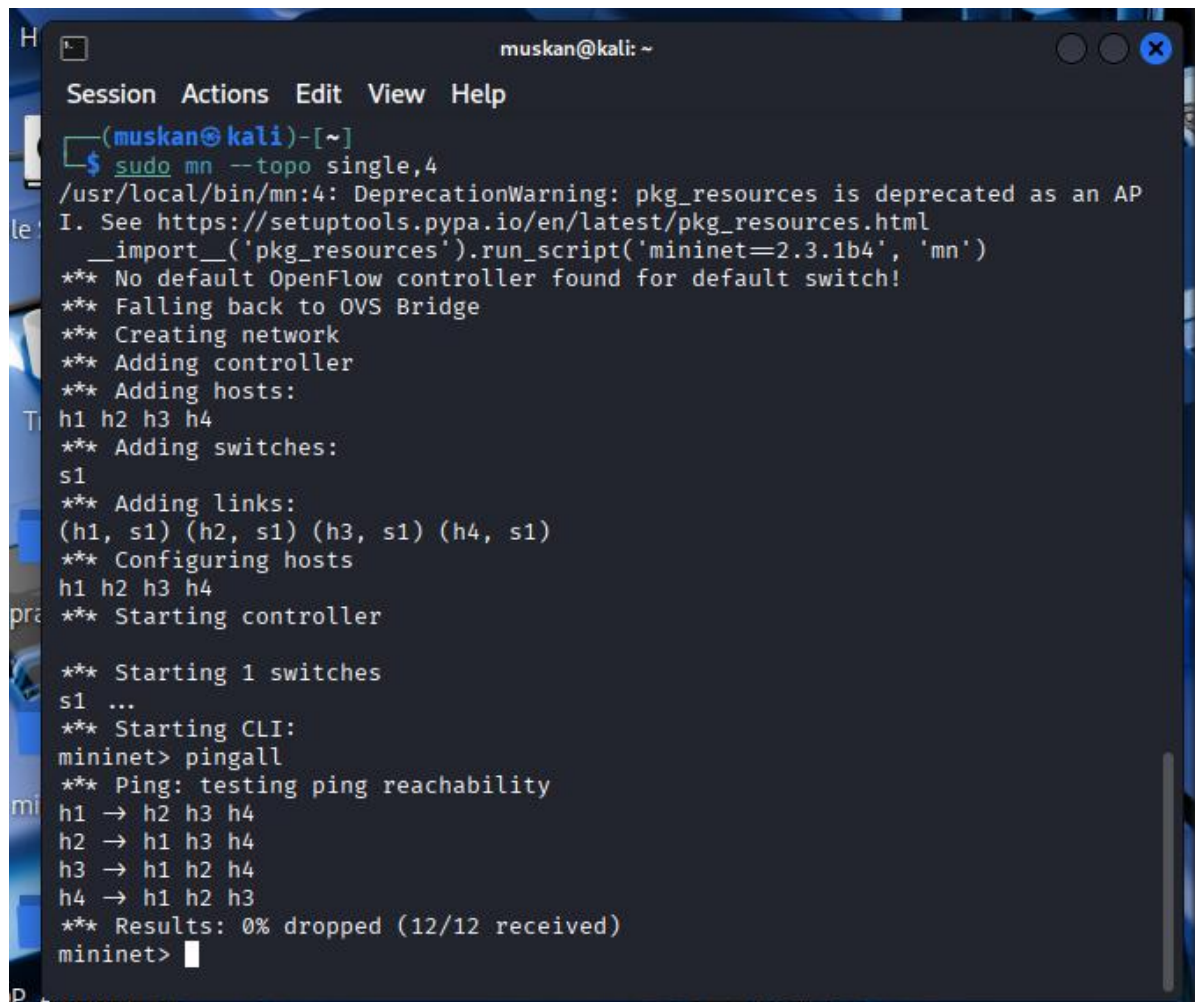
Packet Details (No. 46):

- Ethernet II, Src: KALI LINUX (10:00:00:00:00:00), Dst: VirtualBox (08:00:2b:01:00:00)
- Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.4
- Transmission Control Protocol, Src Port: 5000, Dst Port: 34212, Seq: 0, Ack: 1, Len: 0

Raw Data (Hex):

```
0800 00 04 00 01 00 00 00 20 e0 b6 bb 9c 00 00 00 00 ..... ) .....
0810 0a 00 00 04 13 88 85 a4 ac a4 cc a4 e0 91 a9 fe E - C - 0 0 A .....
0820 a0 12 a9 b0 0c 56 00 00 02 04 05 b4 04 02 08 0a ..... V .....
0830 75 de 75 98 97 64 be 04 01 03 03 00 u u 7 4 .....
```

topology_pingall.png



```
muskan@kali: ~  
Session Actions Edit View Help  
(muskan@kali)-[~]  
$ sudo mn --topo single,4  
/usr/local/bin/mn:4: DeprecationWarning: pkg_resources is deprecated as an AP  
I. See https://setuptools.pypa.io/en/latest/pkg_resources.html  
__import__('pkg_resources').run_script('mininet==2.3.1b4', 'mn')  
*** No default OpenFlow controller found for default switch!  
*** Falling back to OVS Bridge  
*** Creating network  
*** Adding controller  
*** Adding hosts:  
h1 h2 h3 h4  
*** Adding switches:  
s1  
*** Adding links:  
(h1, s1) (h2, s1) (h3, s1) (h4, s1)  
*** Configuring hosts  
h1 h2 h3 h4  
*** Starting controller  
  
*** Starting 1 switches  
s1 ...  
*** Starting CLI:  
mininet> pingall  
*** Ping: testing ping reachability  
h1 -> h2 h3 h4  
h2 -> h1 h3 h4  
h3 -> h1 h2 h4  
h4 -> h1 h2 h3  
*** Results: 0% dropped (12/12 received)  
mininet>
```

wireshark_capture.png

The screenshot displays the Wireshark interface with a network capture of a TCP connection. The packet list on the left shows a series of packets, with packet 26 selected. The packet details pane on the right shows the structure of the selected packet, which is a TCP segment. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
26	71.425561229	10.0.0.2	10.0.0.1	TCP	73	44704 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=83 Len=5 TSval=2920964054 TSecr=
27	71.425761899	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission] 44704 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=83 Len=5
28	71.425764555	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission] 44704 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=83 Len=5
29	71.425766598	10.0.0.2	10.0.0.1	TCP	73	[TCP Retransmission] 44704 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=83 Len=5
30	71.425801012	10.0.0.1	10.0.0.2	TCP	68	5000 → 44704 [ACK] Seq=1 Ack=6 Win=85 Len=0 TSval=2011248158 TSecr=292
31	71.426653948	10.0.0.1	10.0.0.2	TCP	68	[TCP Dup ACK 30w1] 5000 → 44704 [ACK] Seq=1 Ack=6 Win=85 Len=0 TSval=2
32	71.429326525	10.0.0.1	10.0.0.3	TCP	82	5000 → 49706 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14 TSval=48040951 TSecr=
33	71.429347675	10.0.0.1	10.0.0.4	TCP	82	5000 → 36680 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14 TSval=1288707115 TSecr=
34	71.429464901	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission] 5000 → 49706 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
35	71.429467410	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission] 5000 → 49706 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
36	71.429469292	10.0.0.1	10.0.0.3	TCP	82	[TCP Retransmission] 5000 → 49706 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
37	71.429494280	10.0.0.3	10.0.0.1	TCP	68	49706 → 5000 [ACK] Seq=1 Ack=15 Win=83 Len=0 TSval=1755418485 TSecr=48
38	71.429606081	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission] 5000 → 36680 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
39	71.429608042	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission] 5000 → 36680 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
40	71.429608840	10.0.0.1	10.0.0.4	TCP	82	[TCP Retransmission] 5000 → 36680 [PSH, ACK] Seq=1 Ack=1 Win=85 Len=14
41	71.429623523	10.0.0.4	10.0.0.1	TCP	68	36680 → 5000 [ACK] Seq=1 Ack=15 Win=83 Len=0 TSval=3507308962 TSecr=12
42	71.429635680	10.0.0.3	10.0.0.1	TCP	68	[TCP Dup ACK 37w1] 49706 → 5000 [ACK] Seq=1 Ack=15 Win=83 Len=0 TSval=
43	71.429702411	10.0.0.4	10.0.0.1	TCP	68	[TCP Dup ACK 41w1] 36680 → 5000 [ACK] Seq=1 Ack=15 Win=83 Len=0 TSval=
44	68.78.370493121	10.0.0.3	10.0.0.1	TCP	70	49706 → 5000 [PSH, ACK] Seq=1 Ack=15 Win=83 Len=2 TSval=1755425426 TSecr=
45	69.78.370498677	10.0.0.3	10.0.0.1	TCP	70	[TCP Retransmission] 49706 → 5000 [PSH, ACK] Seq=1 Ack=15 Win=83 Len=2

Frame 26: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface
 Linux cooked capture v1
 Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1
 Transmission Control Protocol, Src Port: 44704, Dst Port: 5000, Seq: 1, Ack:
 Data (5 bytes)

Packet 26: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface
 Linux cooked capture v1
 Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1
 Transmission Control Protocol, Src Port: 44704, Dst Port: 5000, Seq: 1, Ack:
 Data (5 bytes)

Packet 26: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface
 Linux cooked capture v1
 Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1
 Transmission Control Protocol, Src Port: 44704, Dst Port: 5000, Seq: 1, Ack:
 Data (5 bytes)